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Demonstrating Large Scale Industrial CCS through CCU – A Case Study for Methanol Production

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Abstract

Methanol is a basic chemical building block used in producing other commodities that we used in our daily lives. In the last decade, due to regulation and policy framework in the transport sector, there is also a significant growth in the use of methanol as fuel (i.e. additives to gasoline, production of MTBE, bio-diesel, synthetic gasoline, etc…).

Methanol is produced by reacting CO and CO_2 with H_2. Traditionally, this is produced in a two-steps process – i.e. syngas production and methanol synthesis. Most of the CO_2 is emitted at the syngas production side of the operation. The question arises if CO_2 is captured, what is its potential for the CO_2 to be used in the methanol synthesis? What is the limitation of CCU on-site? How could this support the deployment and economics of industrial CCS?

This paper aims to present:

- Techno-economic evaluation of deploying CO_2 capture in a mega-methanol plant using NG as feedstock. This plant include a combine reforming process (SMR-ATR configuration) to produce the syngas; methanol synloop and methanol purification unit producing AA grade methanol.
- Concept of on-site CCU as a possible route to expand future methanol production capacity and at the same time promote the deployment of industrial CCS.

Keywords: industrial CCS, on-site CCU, methanol production, syngas production, SMR, ATR, post-combustion CO_2 capture, MEA

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1. Introduction

Methanol (MeOH) is a basic chemical building block used in the production of other commodities. This is an important feedstock in the production of acetic acid, formaldehydes, acrylics, methylamine, dimethyl terephthalate, olefins, and many others. These chemicals are imbedded into several things that we normally use or see in our households or in our daily lives (i.e. paints, furniture, windshield, car parts, carpet, plastic, etc…).

In the last decade, due to regulations and policy framework in the transport sector, there is also a significant growth in the use of methanol as fuel (i.e. blends to gasoline, production of MTBE, DME, bio-diesel, synthetic gasoline, etc…).

Globally, there are more than 100 methanol plants operating worldwide with a cumulative annual production capacity of nearly 100 million metric tonnes (2015 data). Growth in capacity of around 13-15% AAGR is expected in the coming decade. Predominantly, methanol produced outside China are based on NG as feedstocks. Typically, the CO_2 emissions from natural gas based production is around 0.3 – 0.4 tonne/tonne of MeOH.

Methanol is produced mainly from the reaction of CO and CO_2 with H_2. This involves a 2-steps process - i.e. syngas production and methanol synthesis. Most of the CO_2 is emitted from the syngas production side of the operation. The question arises if the CO_2 is to be captured, what is the potential for the captured CO_2 to be used in the methanol plant? What is the limitation of "on-site CCU"? How could this support the deployment and economics of industrial CCS?

This paper attempts to explore these opportunities by evaluating how much CO_2 could be captured in a methanol plant and at what cost. This exercise should provide the baseline information for future studies in the area of industrial CCS and CCU.

Specifically, this paper aims to present:

- Techno-economic evaluation of deploying CO_2 capture in a mega-methanol plant using NG as feedstock.
- Concepts of "on-site CCU" as a possible route to expand future methanol production capacity and at the same time identify synergies with the deployment of industrial CCS.

The first part of this paper presents the techno-economic evaluation of a mega-methanol plant which include a two-step reforming process (SMR-ATR configuration), methanol synloop and methanol purification unit producing 5000 MTPD of AA grade methanol. The methanol plant with CCS captures the CO_2 from the flue gas of the SMR using amine solvent.

The second part of this paper explores how CCU and CCS could work hand in hand within the environment of the methanol plant. The discussion will evolved around the concept of using CO_2 as feedstock to the methanol plant and identify what could be the challenges when integrating CCU with CCS. One of the objectives of this discussion is to raise awareness and understanding of the limitations of the technology when CCU is considered. In particular, this paper would recommend what could be the next step in developing the concept of CCU and in an industrial CCS setting.

Note:

The discussion presented in this paper will be valid only to the production of methanol from NG or other light hydrocarbon as feedstock.

2. Techno-Economics of Capturing CO_2 from a Mega-Methanol Plant using NG as Feedstock (Part 1)

This part of the paper presents the performance and cost of capturing CO_2 from a methanol plant (producing 5000 MTPD of grade AA methanol).

The methanol production process selected for this study is based on a typical process flow scheme that consists of a single train two-step or combined reformers which consist of conventional steam methane reformer (SMR) in series with an oxygen blown autothermal reformer (ATR). The syngas produced is then fed into the methanol synthesis loop (reactor); and the AA grade methanol is recovered from the methanol distillation unit.

2.1. Basis of Design

Methanol Plant

The study evaluates the techno-economic performance of a nominal 5000 t/d methanol plant without and with CO_2 capture using natural gas as feedstock.

The methanol plant consists of the following unit processes:

- Syngas Generation Unit
  - Feedstock Pre-treatment
  - Pre-Reformer
  - Primary Steam Reformer (SMR)
  - Autothermal Reformer (ATR)
  - Syngas Cooler
- Methanol Production Unit
  - Syngas Compression
  - Methanol Synthesis Loop
  - Methanol Distillation Unit
  - Purge Gas Scrubber
  - Hydrogen Recovery Unit
  - Methanol Storage
- Steam and BFW System
- Air Separation Unit
- Demi Water Plant
- Utilities and Balance of Plant (BoP), consisting of:
  - Cooling Water System
  - Flare System
  - Interconnecting
  - Drain System
  - Buildings (CR, Lab, Electrical Sub-station).

For the methanol plant with CO_2 capture, the following process is added:

- Chemical absorption based CO_2 capture plant (using MEA solvent)
- CO_2 compression and dehydration unit

Figures 1 and 2 presents the simplified block flow diagram of the methanol plant without and with CCS.

The description of these processes and the detailed heat and mass balances presented in the IEAGHG main report.

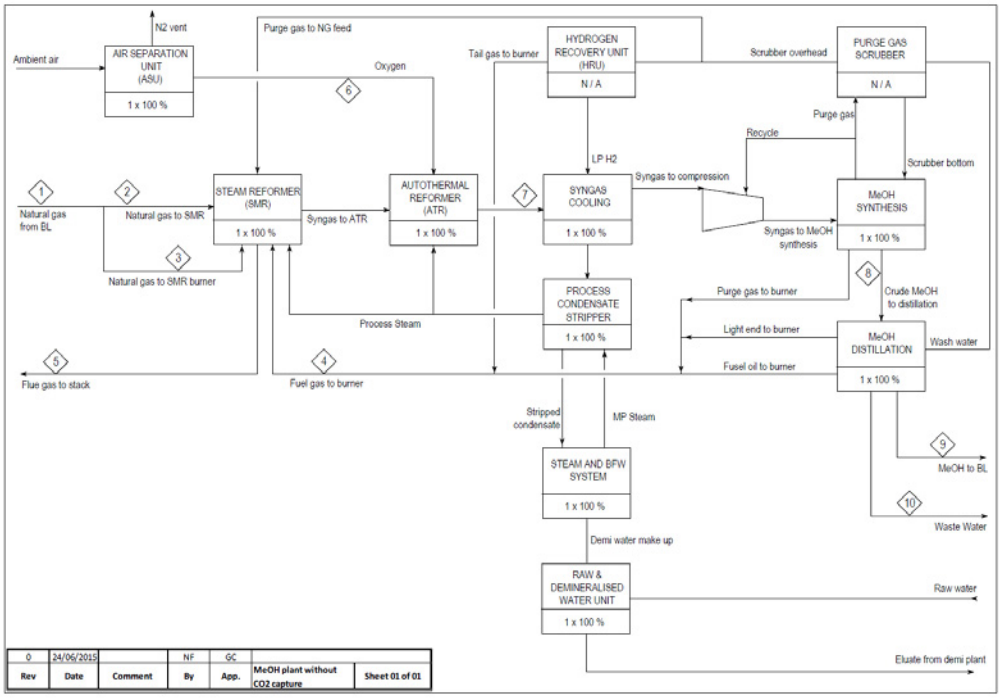


Figure 1: simplified block flow diagram of the methanol plant without CCS

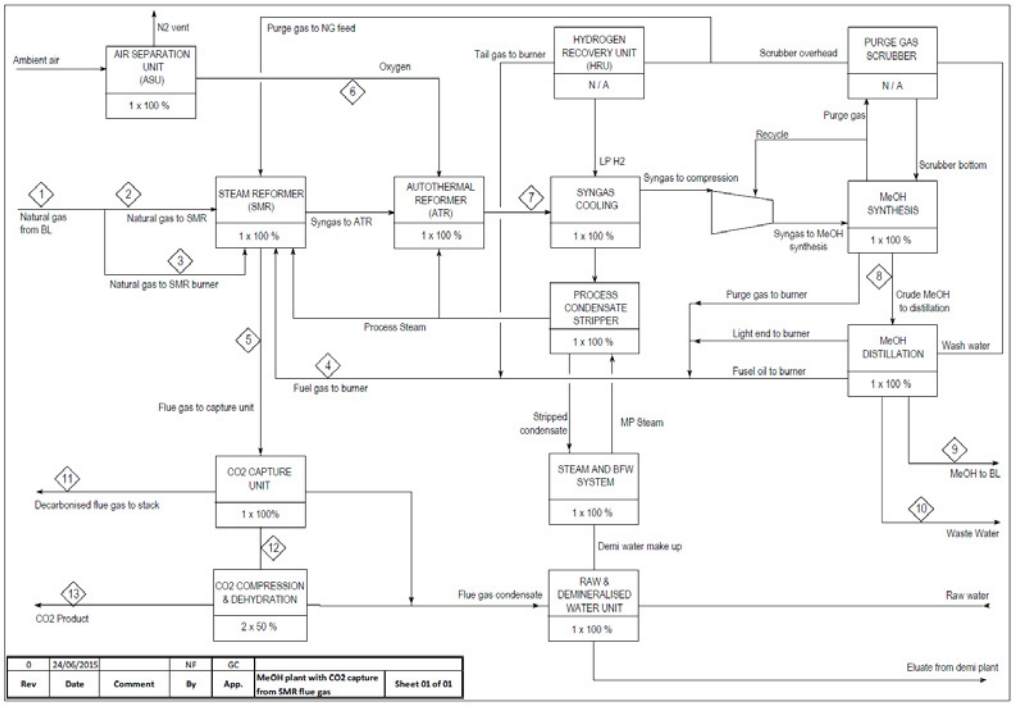


Figure 2: simplified block flow diagram of the methanol plant with CCS

Criteria for Economic Evaluation

Where applicable, the criteria for economic evaluation used in this study is based on the information retrieved from IEAGHG document "Criteria for Technical and Economic Assessment of Plants with Low CO_2 Emissions" Version C-6, March 2014. Other key criteria and assumptions relevant to the operation of the methanol plant are based on the information provided by Amec Foster Wheeler.

The criteria used in the evaluation of the cost of hydrogen production and CO_2 avoided cost are summarized below:

- Plant Life

  The plant is design for an economic life of 25 years.

- Financial leverage (debt / invested capital)

  All capital requirements are treated as debt, i.e. financial leverage equal to 100%.

- Discount Rate

  Discounted cash flow analysis is used to evaluate the levelized cost of H_2 production and CO_2 avoided cost. The discount rate of 8% is assumed.

- Inflation Rate

  Not considered in the discounted cash flow analysis.

- Depreciation

  Not considered in the discounted cash flow analysis. The results presented in this study is reported on Earnings Before Interest, Taxes, Depreciation and Amortization (EBITDA) basis.

- Design and Construction Period

  The design and construction period and the curve of capital expenditure assumed in this study is presented below:

  ⇒ Construction period: 3 years

  ⇒ Curve of capital expenditure

Year	Investment cost %
1	20
2	45
3	35

- Decommissioning Cost

  This is not included in the discounted cash flow analysis. The salvage value of equipment and materials is normally assumed to be equal to the costs of dismantling and site restoration, resulting to a zero net cost for decommissioning.

- Estimate accuracy

  The estimate is based on AACE Class 4 estimate (with accuracy in the range of +35%/-15%), using 4Q2014 price level, in euro (€).

The cost of the methanol plant is estimated based on the capital expenditure and annual operating cost. These are defined as follows:

- Capital Cost

  The capital cost is presented as the Total Plant Cost (TPC) and the Total Capital Requirement (TCR).

  TPC is defined as the installed cost of the plant, including project contingency. This is broken down into:
  - ⇒ Direct materials
  - ⇒ Construction
  - ⇒ EPC services
  - ⇒ Other costs
  - ⇒ Contingency

  TCR is defined as the sum of:
  - ⇒ Total plant cost (TPC)
  - ⇒ Interest during construction
  - ⇒ Owner's costs
  - ⇒ Spare parts cost
  - ⇒ Working capital
  - ⇒ Start-up costs

  For each of the cases the TPC has been determined through a combination of licensor/vendor quotes, the use of Amec Foster Wheeler's in-house database and the development of conceptual estimating models, based on the specific characteristics, materials and design conditions of each item of equipment in the plant. The other components of the TCR have been estimated mainly as percentages of the TPC of the plant.

- Fixed Operating Cost

  The fixed operating cost includes the following:
  - ⇒ direct labor cost
  - ⇒ administrative and general overhead cost
  - ⇒ annual operating and maintenance cost
  - ⇒ insurance
  - ⇒ local taxes and fees

- Variable Operating Cost

  The variable operating cost include the consumptions of the following key items:
  - ⇒ Feedstock (natural gas)
  - ⇒ Raw water make-up
  - ⇒ Catalysts
  - ⇒ Chemicals

The assumptions used to calculate of the total capital requirements:

- Interest during construction

  Interest during construction is calculated from the plant construction schedule and the interest rate is assumed to be the same as the discount rate.

- Owner's cost

  Owner's costs are assumed to be all incurred in the first year of construction, allowing for the fact that some of the costs would be incurred before the start of construction.

  7% of the TPC is assumed to cover the Owner's cost and fees.

- Spare parts cost

  0.5% of the TPC is assumed to cover spare part costs. It is also assumed that spare parts have no value at the end of the plant life due to obsolescence.

- Working capital

  The working capital includes inventories of fuel and chemicals (materials held in storage outside of the process plants). Storage for 30 days at full load is considered for chemicals and consumables. It is assumed that the cost of these materials are recovered at the end of the plant life.

- Start-up cost

  Start-up costs consist of:
  - ⇒ 2% of TPC, to cover modifications to equipment that needed to bring the unit up to full capacity.
  - ⇒ 25% of the full capacity fuel cost for one month, to cover inefficient operation that occurs during the start-up period.
  - ⇒ Three months of operating and maintenance labor costs, to include training.
  - ⇒ One month of chemicals, catalyst and waste disposal costs and maintenance materials costs.

The assumptions used to calculate of the fixed operating cost are as follows:

- Direct labor cost

  This study assumes 56 personnel needed for the Base Case and 5 more additional personnel for all cases with CO_2 capture. The yearly cost of the direct labor is calculated assuming an average salary of 60,000 €/y.

- Admin and general overhead cost (indirect labor cost)

  This study assumes that the indirect labor cost is equal to 30% of the direct labor and maintenance labor cost.
  Generally, admin and general overhead cost is dependent on the company's organization structure and complexity of its operation. This normally covers functions which are not directly involved in the daily operation of the plant.

- Annual operating and maintenance cost

  A precise evaluation of the cost of maintenance would require a breakdown of the costs amongst the numerous components and packages of the plant. Since these costs are all strongly dependent on the type of equipment selected and their statistical maintenance data provided by the selected vendors. This kind of evaluation of the maintenance cost is premature for this type of study.

  For this reason the annual maintenance cost of the plant is estimated as a percentage of the TPC. 1.5% is assumed for all cases. This generally applies to all of the major processes, utilities and off-sites.

  Additionally, estimates can be separately expressed as maintenance labor and maintenance materials. A maintenance labor to materials ratio of 40:60 can be statistically considered for this study.

- Insurance cost

  0.5% of the TPC is assumed to cover the insurance cost.

- Local taxes and fees

  0.5% of the TPC is also assumed to cover the local taxes and fees.

Key assumptions used in estimating the variable operating cost are as follows:

- The assumed prices of the consumables and miscellaneous items are presented in Table 1.

Table 1: Assumed prices for consumables and other miscellaneous items

Item	Unit	Cost	Sensitivity Range
Natural gas	€/GJ (LHV)	6	4 to 16
Raw process water	€/m^3	0.2	-
Electricity	€/MWh	80	40 to 120
CO_2 transport and storage	€/t CO_2 stored	10	0 to 20
CO_2 emission cost	€/t CO_2 emitted	0	0 to 100

- Chemical and Catalyst Cost

  The cost of chemicals is assumed fixed at an annual cost of €200,000 for all cases. This generally accounts for the cost of chemicals used in the treatment of demi-water, process water, boiler feed water, cooling water, and others.

  The catalyst cost which covers the catalyst used in the syngas production and methanol synthesis is also fixed at €3,800,000 per year.

2.2. Definition of Levelised Cost and Cost of CO_2 Avoidance

Levelized Cost of Methanol (LCOMeOH)

The Levelized Cost of Methanol Production (LCOMeOH) is used to calculate the selling price of methanol which enables the present value from all sales of methanol over the economic lifetime of the plant to equal the present value of all costs of building, maintaining and operating the plant over its lifetime. In other word, the selling price of the methanol is calculated based on the assumption that NPV = 0 (over the whole life time of the plant).

The method of calculation is based on a discounted cash flow analysis. This is similar to how the Levelized Cost of Electricity (LCOE) are calculated in other IEAGHG studies, except that it is necessary to take into account the revenues from the sale of electricity as co-product.

The LCOMeOH in this study is calculated assuming constant (in real terms) prices for fuel and other costs and constant operating capacity factors throughout the plant lifetime, apart from lower capacity factors in the first year of operation.

Cost of CO_2 avoidance

Costs of CO_2 avoidance were calculated by comparing the CO_2 emissions per tonne MeOH and the LCOMeOH of plants with capture and a reference plant without capture.

$$CO_2 \text{ Avoidance Cost (CAC)} = \frac{\text{LCOMeOH}_{CCS} - \text{LCOMeOH}_{Reference}}{CO_2\text{Emissions}_{Reference} - CO_2\text{Emissions}_{CCS}}$$

where:

- CAC is expressed in € per tonne of CO_2
- LCOMeOH is expressed in Euro per tonne of MeOH
- CO_2 emission is expressed in tonnes of CO_2 per Nm^3/h MeOH

2.3. Summary of Results

Plant Performance

Table 2 presents the summary of the energy balance of the plant without and with CCS.

For both cases (i.e. methanol plant without and with capture) the total natural gas consumption and the amount of the methanol produced are fixed.

As it could be noted that the main difference in the plant performance could be manifested in the amount of electricity imported by the plant. This has increased by 17.9 Mwe for the methanol plant with CCS (i.e. +96% as compared to the Base Case). This is mainly due to the reduced amount of steam available to the methanol plant to drive various machineries (compressors, pumps, etc…) as most of the excess steam are diverted to the CO_2 capture plant (+11 Mwe) and the electricity demand of the CO2 capture and compression plant (+6.9 Mwe).

Table 3 presents the carbon balance of the plant without and with CCS.

Table 2: Energy Balance of the Methanol Plant

Performance Data			
		Base case	CO_2 capture case
INLET STREAMS			
Natural Gas Feedstock	t/h	119.098	119.098
Natural Gas Fuel	t/h	17.119	17.119
Natural Gas LHV	MJ/kg	46.50	46.50
Total Energy Input	MW	1,760	1,760
OUTLET STREAMS			
Methanol Product to BL	TPD	5,000	5,000
	t/h	208.36	208.36
Methanol LHV	MJ/kg	20.094	20.094
Total Energy in product	MW	1,163	1,163
POWER BALANCE			
Methanol Plant Power consumption	Mwe	11.150	20.295
Steam and BFW consumption	Mwe	2.920	2.920
Utilities + BoP consumption	Mwe	4.400	6.250
CO2 capture plant	Mwe	-	1.655
CO2 Compressor	Mwe	-	5.200
Power import from the grid	Mwe	18.470	36.320
SPECIFIC CONSUMPTIONS			
Natural Gas Feedstock	GJ/ ton MeOH	26.579	26.579
Natural Gas Fuel	GJ/ ton MeOH	3.820	3.820
Feed + Fuel	GJ/ ton MeOH	30.399	30.399
SPECIFIC EMISSIONS			
Specific CO_2 emission MeOH	t/ ton	0.3533	0.0353
Equivalent CO_2 in MeOH product		79.3%	79.3%
Captured CO_2 to storage		NA	18.4%
Overall CO_2 not emitted		79.3%	97.7%

Table 3: Overall Carbon Balance of the Plant

CO_2 removal efficiency	**Equivalent flow of CO_2** kmol/h	
	Methanol plant w/o capture	**Methanol plant with capture from flue gas**
Natural gas feedstock	7168	7168
Natural gas fuel	1030	1030
TOTAL IN	**8199**	8199
Methanol carbon content	6503	6503
CO_2 to storage	-	1505
Total not emitted ©	**6503**	**8008**
Flue gas to stack	1673	167
Vents	23	23
Emission	**1696**	**190**
TOTAL OUT	**8199**	**8199**
Equivalent CO_2 in MeOH product, %	**79.3%**	**79.3%**
Captured CO_2 to storage, %		**18.4%**
Overall Carbon Not Emitted %	**79.3%**	**97.7%**

Economic Evaluation

Tables 4 and 5 present the summary of the total capital cost and annual operating cost of the plant without and with CCS.

Table 4: TPC and TCR of the methanol plant

		Methanol plant w/o CO_2 capture	Methanol plant with CO_2 capture
Methanol plant (including ASU and steam/BFW system)	M€	600	600
Methanol storage	M€	8.7	8.7
Utilities and BoP	M€	85.0	92.8
CO_2 capture unit	M€	-	88.1
CO_2 compression unit	M€	-	20.5
Total Plant Cost (TIC)	**M€**	**693.7**	**810.1**
Contingency	-	20%	20%
Total Plant Cost (TPC)	**M€**	**832.4**	**972.1**
Total Capital Requirement (TCR)	**M€**	**1082.3**	**1264.6**

Table 5: Annual operating cost of methanol plant

O&M COSTS		
	MeOH plant w/o CO2 capture €/year	MeOH plant with CO2 capture from SMR flue gas €/year
Fixed Costs		
Direct labour	3,360,000	3,660,000
Adm./gen overheads	2,506,400	2,847,800
Insurance & Local taxes	8,324,400	9,721,200
Maintenance	12,486,600	14,581,800
Subtotal	**26,677,400**	**30,810,800**
Variable Costs (Availability = 95%)		
Feedstock + Fuel	299,625,600	299,625,600
Electricity	11,649,400	22,907,800
Water Makeup	89,700	54,700
Chemicals and Catalysts	4,000,000	4,000,000
Subtotal	**315,364,700**	**326,588,100**
TOTAL O&M COSTS	**342,042,100**	**357,398,900**

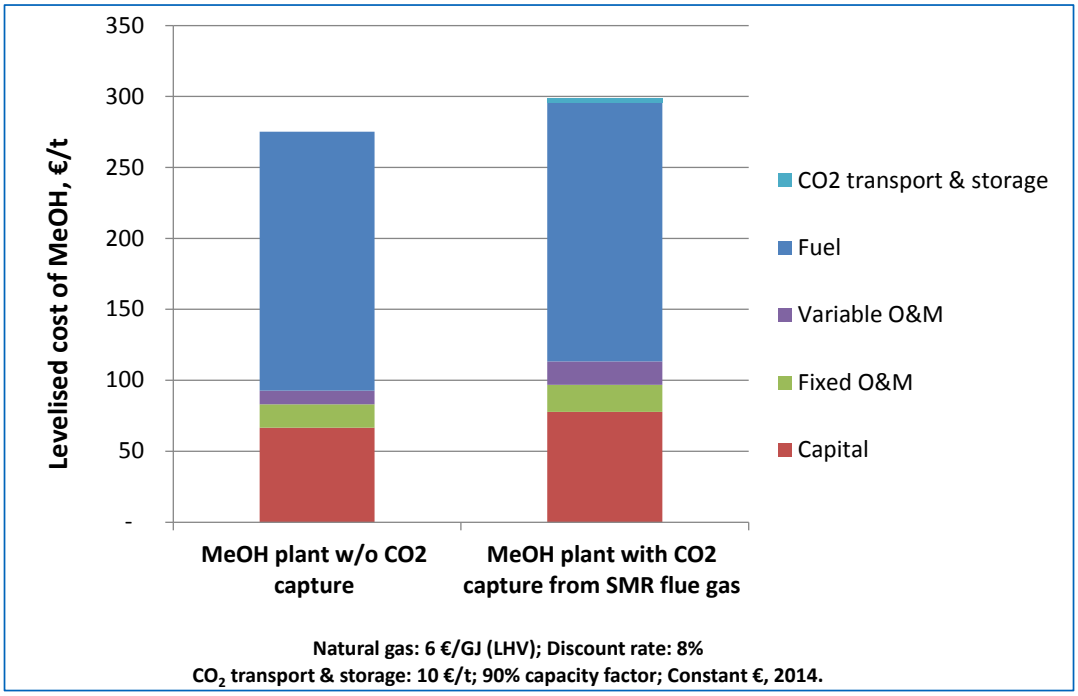


Figure 3 present the levelised cost of methanol production which also indicate the breakdown of the cost.

3. Feasibility of "on-site CCU" and industrial CCS: Case Study on Methanol Plant Setting (Part 2)

This part of the paper presents the concept of "on-site CCU" as a possible route to expand future methanol production capacity and at the same time identify possible synergy with the deployment of industrial CCS.

CO_2 capture and use (CCU) has been widely promoted in various publications and policy discussion. However, there are limitation to CCU in an industrial setting like a methanol plant.

The use of CO_2 to boost production of methanol has been recognized by the major industry players. In fact, this has been deploy as a way to de-bottleneck or boost methanol production. Nonetheless, it is worthwhile to discuss about the concept of "on-site CCU" and to take note that it is not always applicable universally.

This paper specifically aims:

⇒ To explain the different concept on how additional CO_2 could be used within the methanol plant

⇒ To provide awareness on the limitations of the concept of "on-site CCU" within the methanol plant setting.

⇒ To recommend future work to further understand the performance and cost of the methanol plant with "on-site CCU" and industrial CCS.

3.1. Review - Syngas Production

The first step in the production of methanol is the manufacture of the syngas. Figure 4 presents the most common configuration used in industry to produce the syngas from NG. These reformers could cover capacities that are suitable for the production of less than 500 to 10,000 MTPD of methanol. Other types of reforming technologies (such as gas heated reforming) could also be deployed to increase capacity or improve efficiencies of the syngas production.

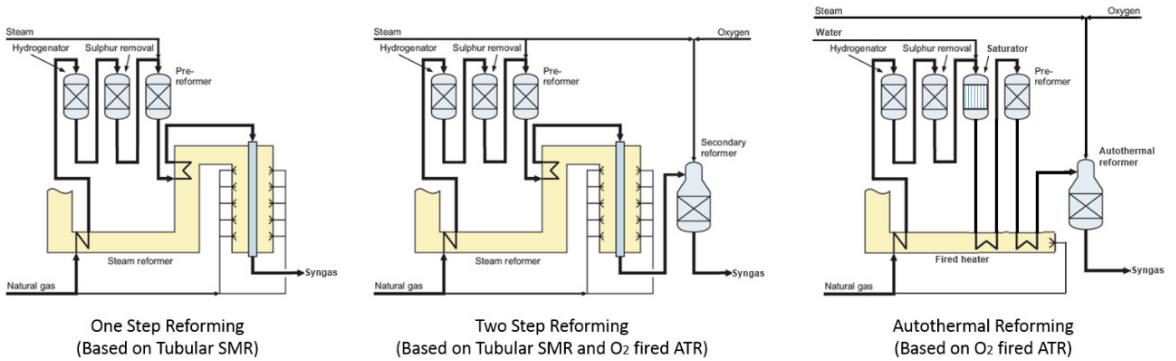


Figure 4: Syngas production – types of reformers (adapted from Haldor Topsoe)

The choice of technology to produce the syngas depends on the scale of operation. A single train tubular steam methane reforming is the cheapest at a capacity below 1,000–1,500 MTPD of MeOH. However, a single train two-step / combined reforming is already common for capacities between 2,500 and 5,000 MTPD of MeOH, whereas a single train auto-thermal reforming is favoured at capacities greater than 6,000 – 7,000 MTPD of MeOH.

Figure 5 illustrates that as the capacity of the methanol plant increases, the cost of the tubular reformer will also increase proportionately, whereas the cost of the air separation unit will increase much less due to the economy of scale. For this reason, the use of oxygen fired autothermal reformer is economical for large scale methanol plants.

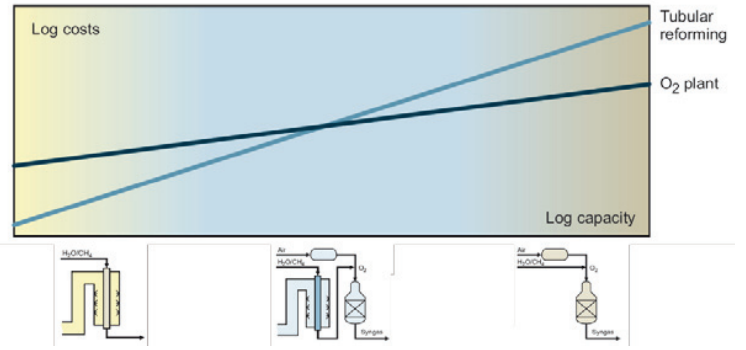


Figure 5: Relative cost of tubular reforming and air separation unit – impact of economy of scale (adapted from Haldor Topsoe)

3.2. Review – Chemistry of Syngas Production and Methanol Synthesis

The key to understanding the use of CO_2 in methanol production is to appreciate the different key reactions involved in the methanol production. Tables 6 and 7 present the key reactions involved in the syngas production and methanol synthesis respectively.

Table 6: Key reactions involved in syngas production

Process	Key Reactions	ΔH_{rxn} (kJ/mol)
Steam Reforming	$CH_4 + H_2O \leftrightarrow CO + 3\ H_2$	206
	$CnHm + n\ H_2O \leftrightarrow n\ CO + (n+m/2)\ H_2$	
	$CO + H_2O \leftrightarrow CO_2 + H_2$	-41
CO_2 Reforming	$CH_4 + CO_2 \leftrightarrow 2\ CO + 2\ H_2$	247
Autothermal Reforming	$CH_4 + 1½\ O_2 \leftrightarrow CO + 2\ H_2O$	-520
	$CH_4 + H_2O \leftrightarrow CO + 3\ H_2$	206
	$CO + H_2O \leftrightarrow CO_2 + H_2$	-41

Table 7: Key reactions involved in methanol synthesis

Process	Key Reactions	ΔH_{rxn} (kJ/mol)
Methanol Synthesis	$CO_2 + 3\ H_2 \leftrightarrow CH_3OH + H_2O$	-40.9 (@ 50 Bar)
	$CO + 2\ H_2 \leftrightarrow CH_3OH + H_2O$	-90.7 (@ 50 Bar)
	$CO_2 + H_2 \leftrightarrow CO + H_2O$	49.8 (@ 50 Bar)
	Other side reactions producing by-products such as ketone, higher alcohol, etc…	

One of the important factors in methanol production is the module number or the stoichiometric ratio between CO_2, CO and H_2. This is defined as:

$$\text{Module No. (M)} = \frac{[H_2] - [CO_2]}{[CO_2] + [CO]} \quad [1]$$

Ideally, the module M should be close to 2. If M > 2, it means that the amount of unreacted H_2 from the methanol synthesis is too much; whilst if M < 2 means that the amount of hydrogen in the syngas is a deficit.

Additionally, the reactivity number as defined by the equation below, is an important parameter in relation to the catalyst activity. Based on traditional copper/chromium based catalyst, it is preferred to have a reactive number greater than 16.

$$\text{Reactivity No.} = \frac{[CO]}{[CO_2]} \quad [2]$$

Typically, the composition of the syngas obtainable from a one-step reformer based on natural gas as feedstock will produce a surplus of H_2 of about 40%. Whilst two step reformer produces syngas with nearly ideal module number and the autothermal reforming producing syngas with sub-stoichiometric ratio.

3.3. Use of CO_2 in Methanol Production

With the use of CO_2 in methanol production, it is worthwhile to highlight that CO_2 could be introduced in the syngas production or in the area of the methanol synthesis.

The introduction of CO_2 in the feedstock of the steam methane reformer promotes CO_2 reforming. However, there is a limitation to how much CO_2 could be introduced. Roughly, this is around 25% of the feedstock.

On the other hand, the introduction of CO_2 in the methanol synthesis could be limited by the catalyst used in the methanol synthesis. Higher reactivity catalyst is needed to compensate for the lower reactivity number. Furthermore, the introduction of CO_2 in the methanol synthesis could also be limited by the type of reactor used. Adiabatic reactor (although simple and cheaper, but more complex to operate) would have less flexibility to handle additional CO_2 than a BWR or steam raising type reactors.

3.4. Engineering Considerations of Using CO_2 in the Methanol Production

The use of CO_2 to boost the production of methanol has been used in several plants in Middle East since 2004 (i.e. Iran and Saudi Arabia). Several of these plants make use of excess CO_2 from an ammonia complex.

However, it is important to spell out what are the limitations to the use of CO_2 in the current fleet of methanol production.

The engineering considerations of using additional CO_2 to boost methanol production should be considered in two different perspective based on revamp (retrofit) or new build options. For revamp option, the use of CO_2 is only suitable to one-step reforming. However, for new build option, some adjustment should be necessary in the operation of the two step reforming before it could be suitable. For ATR based methanol plant, use of CO_2 is most unlikely due to the deficit of hydrogen unless additional reformer in parallel or in series to the ATR is added.

Figure 6 illustrates the possible gains of using CO_2 to boost methanol production in one-step reforming (for a retrofit case). Figures 7 and 8 are some of the schemes that could be considered if CO_2 will be added to the methanol synthesis section of the plant.

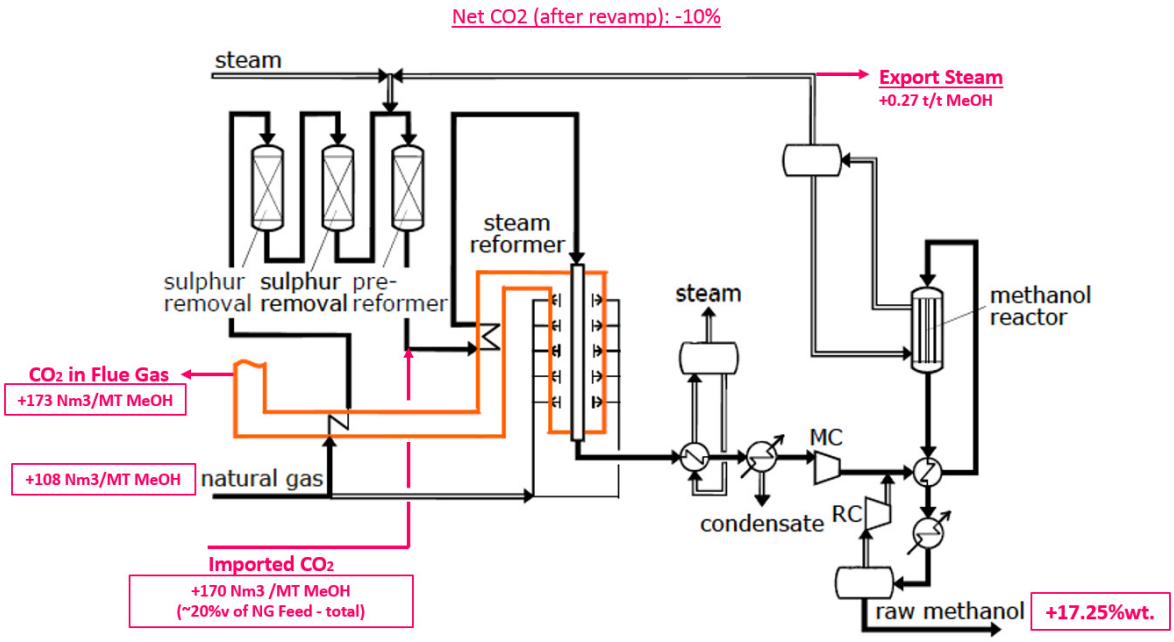


Figure 6: Impact of boosting methanol production by addition of CO_2 in a one-step reforming methanol plant (revamp option)

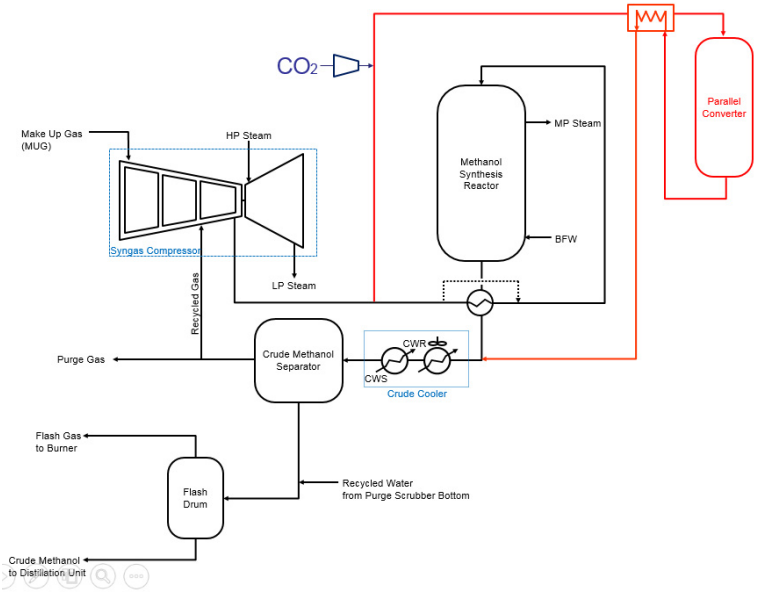


Figure 7: Boosting of methanol production by addition of parallel converter (revamp option for one step reforming)

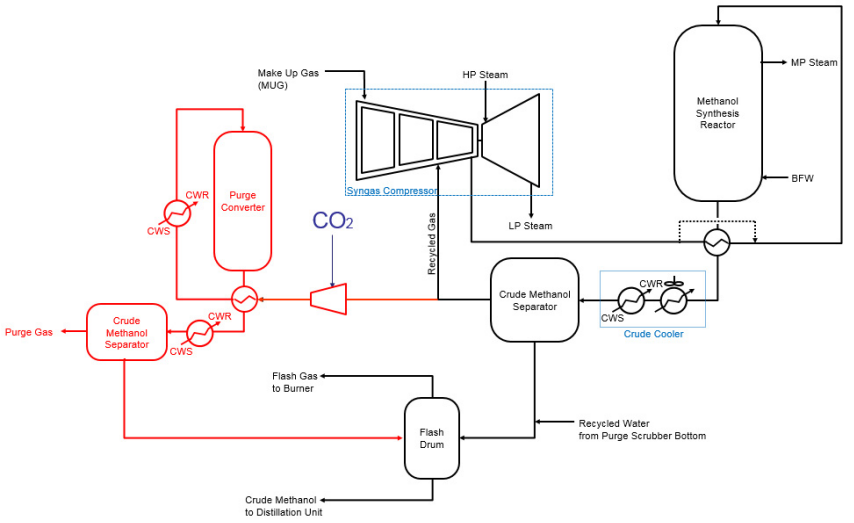


Figure 8: Boosting of methanol production by addition of purge converter (revamp option for one step reforming)

For revamp case, the following options for the addition of the CO_2 could be considered to accommodate the increase of 20% capacity:

⇒ Use of more active methanol synthesis catalyst (suitable for handling higher volume of CO_2)
⇒ Use of additional converter (depending on the type of reactor)
  a. For BWR type reactor – additional converter in parallel or in series are possible
  b. For Adiabatic type reactor – will be limited to converter in series after the main reactor
⇒ Use of pre-converter reactor could also be added (handling mainly the make-up gas) – this will be limited to BWR type reactor
⇒ Use of purge converter

For new build plant, similar strategy as discussed above could be used for one-step reforming methanol plant. However, for two-step reforming methanol, an important consideration is on how to increase the module number to accommodate the additional CO_2 to be handled. Critical to this type of plant is on how to manage the natural gas feed to the steam reformer and the secondary reformer.

4. Concluding Remarks

For revamp of the methanol plant – the use of CO_2 to boost production is limited to one-step reforming technology. This could boost methanol production by up to 20% and reduce energy consumption by up to 5%. CCU in this case could support CCS by consuming around 50% CO_2 that could be captured from the flue gas. This should reduce the net amount of CO_2 to be exported to the pipeline.

For new build option, the use of CO_2 is very applicable to one-step reforming.

For two-step reforming, it may depend on how H_2 could be boosted – given that 2 step-reforming scheme has been operated with an ideal module number of 2. Therefore, it is necessary to assess how to accommodate the additional CO_2.

For ATR type of methanol plant, the use of CO_2 is not relevant unless, additional reformer will be added to boost hydrogen content in the syngas.